Automated Plant Leaf Disease Detection using CNN

A Human-Centered Deep Learning Approach

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INTRODUCTION

Plant diseases reduce crop yield and quality

Climate change increases disease spread

Manual detection is:

- Time-consuming
- Error-prone
- Requires expertise

Leaf diseases often appear as spots, discoloration, and texture changes.

SECONDARY RESEARCH – DATASET & INSIGHTS

Studied PlantVillage dataset (~54,000 images)

Covers multiple crops and disease classes

Observations:

- High accuracy in controlled environments
- Performance drops in real-world condition

Background noise, lighting, and leaf orientation affect model accuracy

SECONDARY RESEARCH – CNN MODELS

Studied deep learning models:

ResNet50

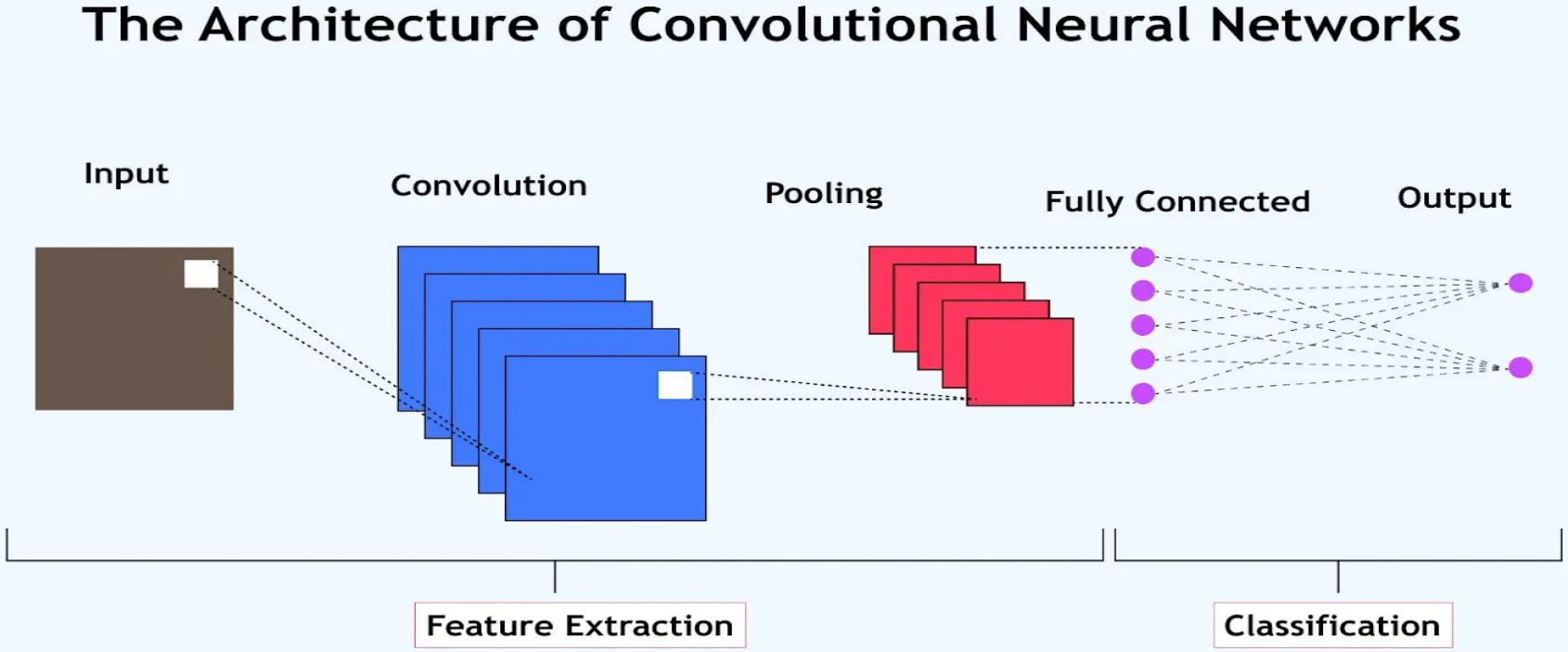
- Uses residual connections
- Solves vanishing gradient problem

EfficientNet

- Optimized architecture
- High accuracy with fewer parameters

Attention Mechanisms

- Focus on important regions
- Ignore background noise



CNN models used for feature extraction and classification

PRIMARY RESEARCH – FIELD STUDY

Conducted real-world observations of plant leaves

Performed:

- Contextual inquiry
- Expert discussions
- Surveys

Identified key problems:

- Difficulty in early disease detection
- Misidentification of symptoms
- Limited access to experts

Example of infected leaf with visible disease spots

PRIMARY RESEARCH – KEY FINDINGS

Disease symptoms vary across plants:

- Spots
- Discoloration
- Leaf damage

Environmental challenges:

- Lighting variation
- Wind (leaf movement)
- Background clutter

Variation in disease patterns and leaf discoloration

ANALYSIS – PROBLEM IDENTIFICATION

CNN models perform well in controlled environments

Real-world images introduce:

- Noise
- Complex backgrounds
- Multiple disease regions

Complex disease patterns under real-world conditions

ANALYSIS – CHALLENGES & SOLUTION

Challenges:

- Overfitting to background features
- Difficulty detecting small lesions
- Lack of interpretability

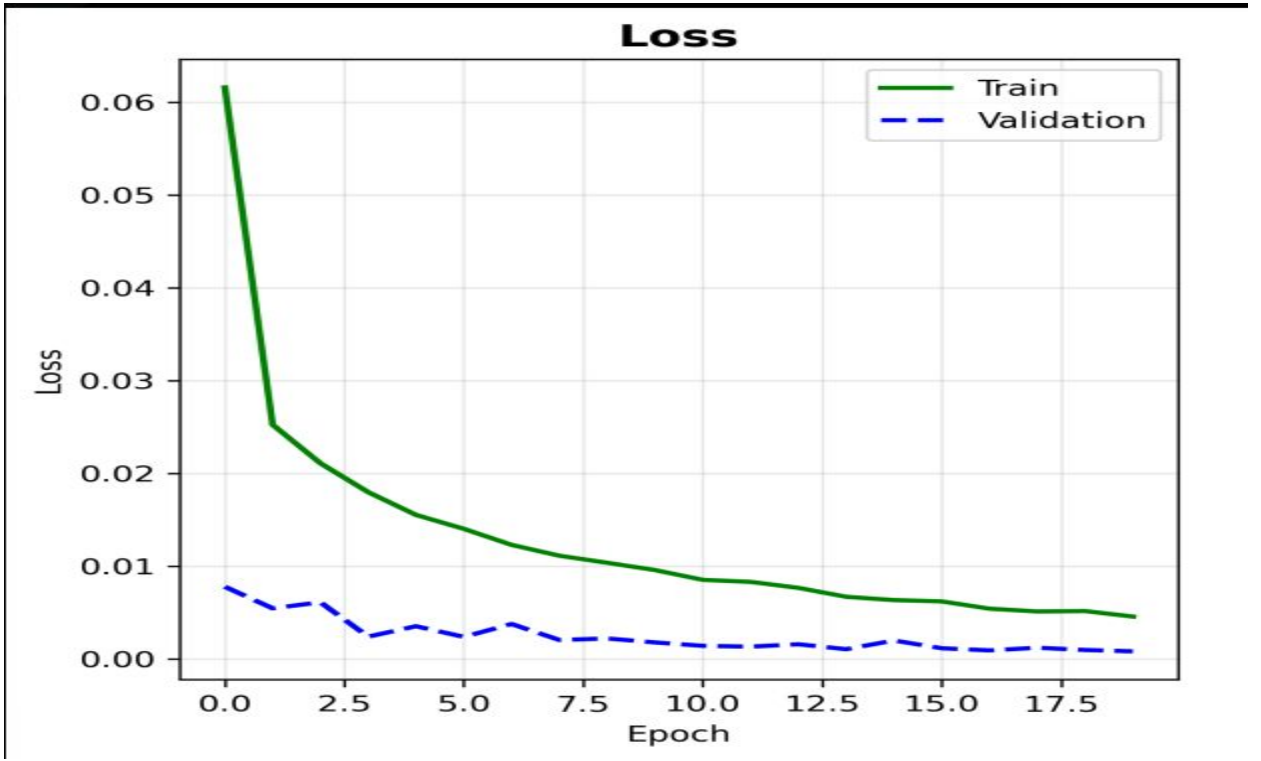
Proposed Solution:

- Use EfficientNet & ResNet50
- Apply Attention Mechanisms
- Use Early Stopping

Irregular disease distribution requiring focused detection

MODEL PERFORMANCE – LOSS ANALYSIS

- Training loss decreases steadily over epochs
- Validation loss remains low and stable



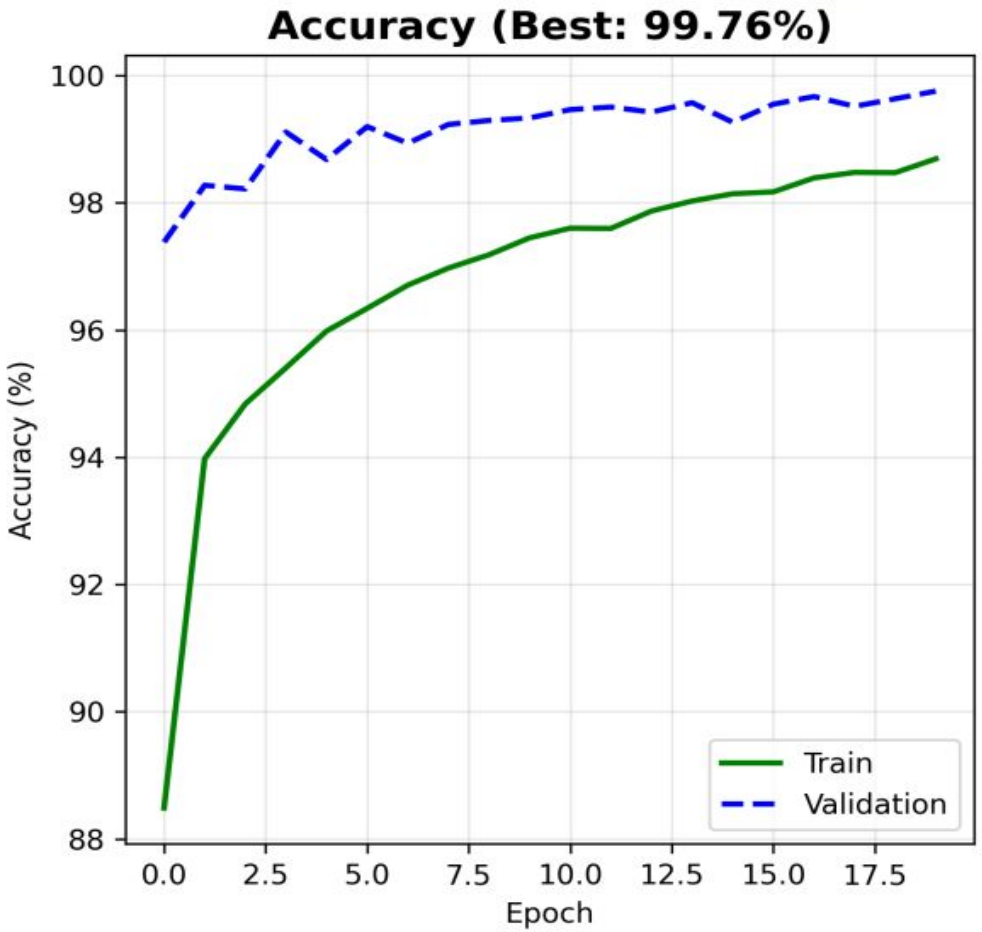
Training vs Validation Loss showing stable convergence

MODEL PERFORMANCE – ACCURACY

Training accuracy increases consistently

Validation accuracy reaches ~99.7%

Training vs Validation Accuracy showing high performance

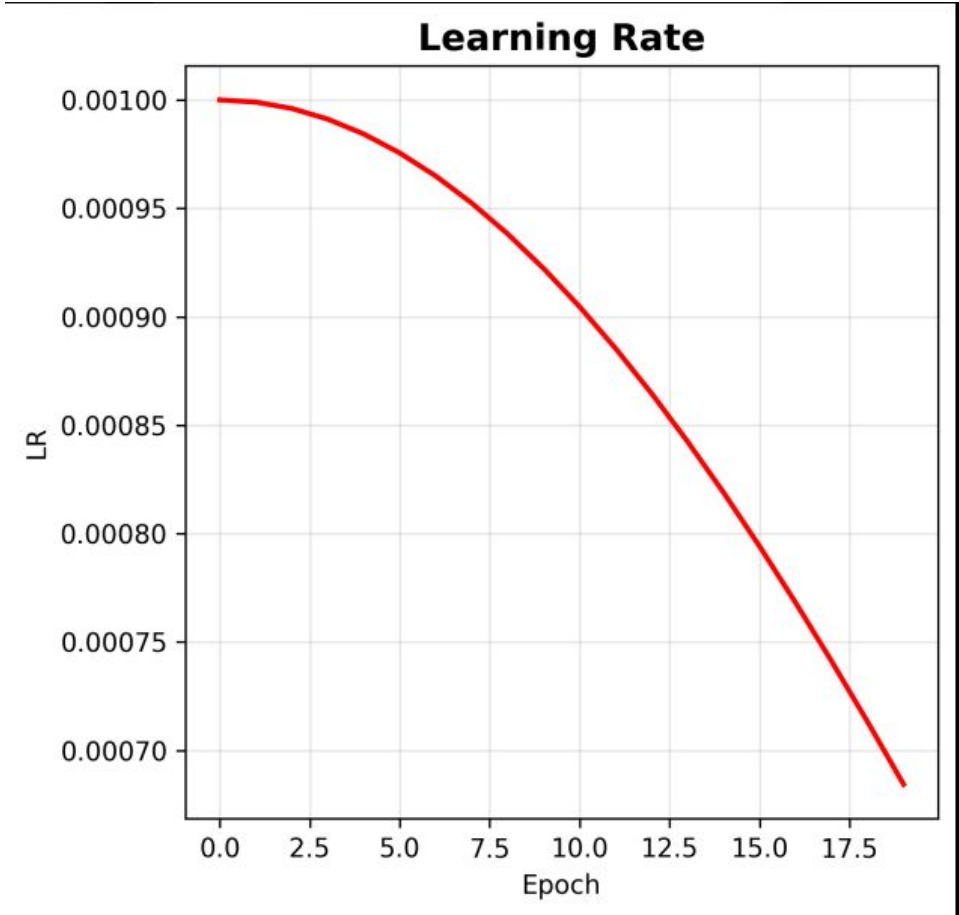


MODEL OPTIMIZATION – LEARNING RATE

Learning rate decreases gradually during training

Helps achieve stable convergence

Learning rate scheduling for optimized training



FINAL RESULTS

Achieved high accuracy (~99.7%)

Training and validation loss are low

No significant overfitting observed

Effective for real-world plant disease detection

REFERENCES

PlantVillage Dataset (Kaggle)

Research papers on CNN for plant disease detection

Deep Learning frameworks (TensorFlow / PyTorch)

YouTube – Design Thinking & CNN tutorials

Scientific articles (Frontiers, MDPI, IEEE)